

8. New minor issues I should have caught:

- Table 15 lists "Qwen2.5-34B-Instruct" with size 3B — almost certainly a typo for Qwen2.5-3B-Instruct.
- Table 15 lists "Mistral-8B-Instruct" — not a standard Mistral release (7B and Nemo-12B are).
- The text says the census spans "0.5B to 32B," but Table 15's smallest model is SmoLLM2-1.7B. Inconsistent.

Corrected Review

ICLR 2027 Review — Anonymous Submission

Title: One Token Owns the Calibration Hessian: Why Layer-wise Quantization Silently Breaks Instruction Following

Summary

The paper identifies and explains a failure mode of GPTQ-style layer-wise post-training quantization on instruction-tuned LLMs. Under a "short-document" calibration protocol (tokenizing 128 C4 documents separately, as GPTQModel and LLM Compressor do by default), 3-bit GPTQ collapses instruction following on Llama-3.1-8B-Instruct (IFEval $0.768 \rightarrow 0.155$) and Qwen2.5-14B-Instruct ($0.820 \rightarrow 0.426$), dropping below round-to-nearest (RTN) at the same bit-width (41 and 27 points below RTN on IFEval, respectively), while MMLU and perplexity remain at or above RTN levels. The authors trace the failure to the MLP down-projection of a layer where the attention sink forms. At this matrix, the calibration Hessian $H = \sum x_t x_t^T$ is effectively rank-one and dominated by the BOS token ($A_{\text{BOS}} \approx 0.998$), while the downstream loss sensitivity $g_t = \|\partial L / \partial y_t\|^2$ concentrates on chat-template tokens ($G_{\text{BOS}} \leq 0.008$). GPTQ's compensation step therefore protects BOS and redistributes its rounding error onto the loss-relevant template positions. The authors derive a closed-form characterization of this redistribution under a dominant token (Theorem 1), predict its direction/row

concentration/amplitude (Corollary 1), localize the resulting "lesion" to one or two matrices by excision and transplantation, and propose a corrected objective that reweights the Hessian by $w_t = g_t / \|x_t\|^2$, which repairs all 3-bit collapses found. The account is supported by a census of 18 quantized instruction-tuned models and a pre-registration log that includes 30 refuted predictions. (The authors also disclose and correct a bug in an earlier version of their layer-wise loop, Appendix B.2, and re-run all affected arms.)

Strengths

1. **The phenomenon is real, important, and well-verified.** The gap between GPTQ and RTN at 3 bits under a default calibration shape is large (41 and 27 points on IFEval for Llama and Qwen, and 27–64 points across generation benchmarks), reproducible across draws, and the failure is genuinely silent on perplexity and MMLU. This alone is a valuable empirical finding for practitioners.
2. **The mechanism is derived, not narrated.** Theorem 1 is a clean Sherman–Morrison-based characterization of the compensation step when one token dominates the curvature. The isotropic corollary (Eq. 7) is interpretable, and the factorization in Corollary 1(iii) (share of BOS error cancelled $\times$ concentration factor) is a nontrivial, quantitative prediction. The theory is local but tight and exact for the steps it covers.
3. **The causal evidence is unusually strong.** Excision (re-rounding one matrix to RTN without extra bits restores IFEval from 0.155 to 0.622), transplantation (the lesion collapses an RTN network, a full-precision network barely notices), subset enumeration to isolate a minimal collapsing set, and activation patching of hidden states after the lesion layer all converge on the same object. This is well beyond the usual ablation.
4. **The pre-registration and negative-results appendix are exemplary.** Thirty refuted predictions, including the paper's own favored hypotheses (a κ threshold, lesion-size ranking, BOS-share cross-model criterion), with explicit statements of what each refutation removed from the account. This is rare and raises confidence in the surviving claims.
5. **The repair is derived from the same theory, not bolted on.** Proposition 2(a) (position-independent output factors cannot change the column-wise update) is a genuine

negative result that explains why K-FAC-style output-side factors do not help, and the $w_t = g_t / \|x_t\|^2$ reweighting follows directly from Proposition 2(b). The ablation showing the two halves fail in complementary places is consistent with the theory.

6. **Generality is checked, not assumed.** 28 models for the premise, 18 for collapse, 6 quantizers (plus RTN baseline), 7 calibration protocols. The boundary is stated cleanly (all 26 with a sink show the mismatch, neither gemma-2 model does).
7. **Honest limitations.** The absorption of the lesion by later layers is explicitly not derived; the Mistral phenotype is localized to two layers but not to a matrix; the 4-bit story is partial. These are stated rather than hidden.
8. **The loop-correction disclosure (Appendix B.2).** The authors document a bug in an earlier version of their layer-wise loop (passing a KV cache into per-layer calls), re-run all affected arms, and report the two results that moved. This raises confidence in the reported numbers.

Weaknesses

Major

W1. The theoretical object (Corollary 1) is derived for an isotropic $M_j = \lambda I$ that is not the regime of interest, and the general case is handled by a hand-wave.

Corollary 1 is stated under $M_j = \lambda I$, which requires the ordinary-token curvature to be isotropic. The paper acknowledges this ("For general M_j , the per-step directions differ") and reports that the per-step direction becomes $v = M_j^{-1}x_0$, but the three quantitative signatures — row concentration along x_0 , readout by $x_0^T x_t$, and the amplitude factorization — are all isotropic results. The measured "sending pattern" cosine with x_0 is 0.52–0.93 across 14/17 sink matrices, i.e., *substantial but variable*, and the cosine with v is 0.02–0.70, i.e., *worse*. This is awkward: the derived direction for the general case fits the data worse than the isotropic approximation it is supposed to generalize. The paper should either (a) argue that M_j is approximately isotropic at the sink-forming matrices and support that with a measurement (e.g., the condition number or the eigenvalue spectrum of M_j), or (b) derive the signatures directly in the general case rather than substituting v post hoc. As written, Theorem 1(a,b) is exact and the corollary is a special case, but the empirical section repeatedly uses the corollary's language ("follows the

sink pattern x_0 ", "overlap with x_0 ") when the actual derived quantity is v . This needs to be tightened.

W2. The readout prediction of Corollary 1(ii) fails on Llama, and the paper's replacement (the row readout $\delta^{(r)}x_t$) is not derived from Theorem 1.

The paper is admirably candid that the isotropic readout $x_0^T x_t$ places template tokens on Qwen (recall 0.52) but not on Llama (recall 0.11), and the whitened overlap $x_0^T M^{-1} x_t$ is worst on Qwen (0.33 vs 0.52 isotropic and 0.55 row) and second-worst on Llama (0.22 vs 0.11 isotropic and 0.54 row). The fallback is the row readout $\delta^{(r)}x_t$, which does place them (recall 0.54 on Llama, 0.55 on Qwen). But $\delta^{(r)}$ is a *row of the quantization residual*, i.e., it depends on the data, not just on the theory. The paper reports that this readout tracks the block error with Pearson 0.99 (Fig. 4c), which is impressive, but "the row that happens to be worst also happens to have the largest readout on the worst tokens" is closer to a post-hoc diagnostic than to a prediction of Theorem 1. A reviewer will ask: is the row readout *predicted* by the theory, or is it the theory's failure mode that you then measured and found correlates with damage? The paper should state this distinction explicitly and, ideally, show that the row readout is derivable from Eq. (6) rather than presented as one of three candidate readouts that happened to work.

W3. "Collapse" is a sharp binary in the paper, but the underlying quantity appears to be continuous, and no criterion is offered.

Section 4 and Section 7 report collapse/no-collapse as a binary (2 of 18 under short documents, 1 of 18 under c4win), and the paper explicitly rules out κ , lesion size, BOS-share, and calibration composition as predictors (all pre-registered and refuted). What remains is "the lesion and the failure of its neighbors to absorb it," with absorption not derived and "on Qwen not yet localized." This is honest but leaves the central phenomenon without a quantitative criterion. A reviewer will press: *given a new model and a new calibration set, can I predict whether it will collapse?* If the answer is no, the paper's practical value is reduced to "validate on free generation and use the corrected objective if you see a sink-dominated early down-projection" — which is a fine recommendation, but weaker than the paper's framing suggests. The paper should either provide a partial predictor (e.g., a bound on the absorption capacity of the later layers) or state plainly that no criterion is currently available and that this is the main open problem.

W4. The corrected objective's costs are under-discussed relative to its benefits.

The corrected objective lowers Llama's MMLU (0.524 → 0.514, *below* the collapsed checkpoint), recovers GSM8K only partially (0.548 vs 0.632 for re-rounding), and costs several points on gemma-2-9b (which has no sink, -4.6 vs RTN). On Qwen it is a few points behind re-rounding on IFEval (0.736 vs 0.768). The paper is candid about these numbers, but the framing in the abstract and introduction ("repairs every collapse we found") understates that the repair is not free and is not a new default. More importantly: **re-rounding the single culprit matrix to RTN achieves 0.622 on Llama and 0.768 on Qwen, i.e., it is competitive with or better than the corrected objective on IFEval, and it costs nothing.** The paper acknowledges that re-rounding "needs the diagnosis of Section 4," but the diagnosis is exactly what the paper provides. A practitioner reading this paper could reasonably conclude that the actionable recommendation is "diagnose the sink matrix and re-round it to RTN," not "adopt a new objective that costs a backward pass, 49% more memory at 8B, and a point of MMLU." The paper needs to make the case for *why* the corrected objective is preferable to the much cheaper re-rounding intervention, beyond "it repairs every collapse we found" (which re-rounding also does, once the matrix is identified). The position prior (constant weight on first eight positions) is even cheaper and reaches 0.623/0.602 on Llama and 0.405/0.418 on Mistral; this deserves more prominence if the goal is practical impact.

W5. The "lesion" framing may overstate the object's status as a discrete, localized entity.

The paper defines the lesion as "its concentrated part" (Section 4) and identifies a minimal collapsing set of one or two matrices. But the excision experiments show a graded picture: re-rounding the layer-1 down-projection alone gives 0.622, re-rounding both layers 0 and 1 gives 0.632, and re-rounding layer 0 alone gives 0.595 (vs. RTN 0.565). These are all partial recoveries, not a return to RTN. The "lesion" is thus not a discrete faulty component whose removal restores function; it is the dominant contributor to a distributed error. The transplantation results (the lesion collapses an RTN network, but the same lesion in a full-precision network costs almost nothing) are the strongest evidence for localization, but they coexist with the finding that "the neighbors decide" absorption. A reviewer may feel that the language of excision/transplant/lesion imports a surgical metaphor that the data only partly supports. This is a framing issue rather than a correctness issue, but it affects how the contribution is perceived.

Minor

m1. The BOS-token double-count. Appendix B.4 notes that on Llama and Mistral the chat template writes a BOS token and the tokenizer adds another, so every prompt carries two BOS tokens, and the position indices of Section 4 count both. This is a potential confound for a paper whose central object is "the BOS token." The single-BOS control (fp16 Llama 0.768 $\rightarrow$ 0.808; collapsed 0.155 $\rightarrow$ 0.158) suggests the conclusion is robust, but a reader will want to know: does the second BOS token change A_{BOS} , κ , or the lesion's location? The paper should either add a two-line measurement or state explicitly that the second BOS is not a sink and does not affect the sink-forming matrix's statistics.

m2. The "short-document" protocol's fidelity to library defaults. Table 1 states that the short-document protocol reproduces GPTQModel's sampling shape with 128 rows, but GPTQModel's README example uses 1,024 rows. With 1,024 rows, Llama gives 0.148 (also collapsed) and Qwen 0.435 (also collapsed), so the phenomenon survives. But the paper's framing ("the calibration protocol that a widely used library applies by default") slightly overstates: the library's *sampling shape* is the default, but the *number of rows* in the paper's main arm is not the library's README default. This should be stated precisely in the main text, not only in Appendix B.1.

m3. Mistral's phenotype is unresolved and the paper says so, but it is load-bearing for the generality claim. Mistral collapses under C4 windows, not short documents; the collapse needs the fourteen matrices of layers 0–1 but neither down-projection alone. This is presented as a limitation, but it also means the "lesion" account does not cover one of the three main models. The generality section would be stronger if it either (a) resolved the Mistral case or (b) reframed the contribution as covering the GPTQ-on-sink-dominated-down-projection phenotype, with Mistral as a distinct phenotype that shares the objective-mismatch premise but not the localization.

m4. The corrected objective's damping interaction. Appendix B.3 notes that because $\lambda = \rho \cdot \text{mean}(\text{diag } H)$ and H' no longer contains the BOS mass, λ falls by the same factor, so ρ was not retuned. This is a reasonable choice, but it means the corrected objective differs from GPTQ in *two* ways (token weights and effective damping), and the ablation does not separate them. A reader could ask whether the repair is partly due to the damping change. The ρ -sensitivity table (0.572 at $\rho=0.01$, 0.638 at 0.05, 0.610 at 0.2)

suggests not, but the ablation should include a "token weights with λ held at the GPTQ value" arm to close this.

m5. The 4-bit results are thin. Section 6 reports that 4-bit GPTQ does not beat 4-bit RTN on IFEval (mean -0.1 , negative on 11/18), and the corrected objective recovers Mistral (+6.3) and Hermes-3 (+5.9). This is interesting but underdeveloped: if 4-bit GPTQ is on average *worse* than RTN, that is a stronger claim than "the 3-bit collapse is silent," and it deserves either more space or a clearer statement of scope.

m6. Figure and table density. The paper has 16 tables and 4 figures across the main text and appendices, many of which are dense number dumps. For an ICLR submission with a 9-page limit, the main text may be better served by moving some tables to the appendix and promoting a single summary figure.

m7. The "AI use statement" disclosure. The statement says generative AI tools were used to "draft and refactor analysis code, to search the literature and to edit the text and figures." ICLR's policy on AI-generated text in submissions is evolving; the authors should verify that this level of disclosure is compliant with the 2027 policy and, in particular, that no content in the paper was *generated* rather than edited.

m8. Inconsistencies in Table 15 and the census description.

- Table 15 lists "Qwen2.5-34B-Instruct" with size 3B — almost certainly a typo for Qwen2.5-3B-Instruct.
- Table 15 lists "Mistral-8B-Instruct," which does not correspond to a standard Mistral release (Mistral-7B and Mistral-Nemo-12B are the known variants); please verify.
- The text says the census spans "0.5B to 32B," but Table 15's smallest model is SmoLLM2-1.7B. Please reconcile.

Questions for the Authors

1. **On W1:** Can you report the eigenvalue spectrum or condition number of M_j at the sink-forming matrices, to justify the isotropic approximation used in Corollary 1? If M_j is strongly anisotropic, does the general-case direction $v = M_j^{-1}x_0$ do better or worse than x_0 at predicting the row structure, and by how much?

2. **On W2:** Is the row readout $\delta^{(r)\top} \mathbf{x}_t$ derivable from Eq. (6), or is it a post-hoc diagnostic? If derivable, please show the derivation; if post-hoc, please state this explicitly and report whether any *a priori* readout (from the theory) achieves comparable recall on Llama.
3. **On W3:** Given a new model and calibration set, is there *any* partial predictor of collapse? For example, does the ratio of the lesion's template-position error to the later layers' capacity to absorb it (however you would operationalize that) correlate with collapse across the 18-model census? If no predictor exists, please state this as the central open problem.
4. **On W4:** Why should a practitioner prefer the corrected objective over re-rounding the diagnosed sink matrix to RTN, given that re-rounding is cheaper, requires no backward pass, and achieves comparable or better IFEval on Llama and Qwen? Is there a setting where re-rounding fails and the corrected objective succeeds?
5. **On m1:** Does the second BOS token (from tokenizer + chat template) affect A_BOS, κ , or the identity of the sink-forming matrix? Please report the single-BOS statistics at the sink matrix.
6. **On m2:** What is the exact row count in the main short-document arm, and how does it compare to the library defaults you cite? Please state this in the main text.
7. **On m4:** Can you add an ablation of the corrected objective with λ held at the GPTQ value, to separate the effect of token weights from the effect of effective damping?
8. **On the pre-registration:** The log is described as released with the submission. Will reviewers have access to it during review, or only at camera-ready? If the former, please ensure the link is in the reproducibility statement (currently a TODO).
9. **On m8:** Please reconcile the model names and size range in Table 15 with the text.

Soundness

3 / 4 — The empirical work is strong, the causal evidence (excision, transplantation, subset enumeration, activation patching) is unusually thorough, and the theory is correct for the steps it covers. The score is held back by the gap between the isotropic corollary and the general case (W1), the post-hoc status of the row readout (W2), and the absence of a collapse criterion despite extensive search (W3).

Presentation

3 / 4 — The paper is dense but generally well-organized, and the prose is clear. The heavy reliance on tables of numbers is a stylistic choice. The main text would benefit from moving some tables to the appendix and promoting a single summary figure. The abstract and introduction are well-written but slightly oversell the repair (W4).

Contribution

3 / 4 — The identification of the failure mode is a genuine contribution to the PTQ literature, and the mechanism (objective mismatch → error misallocation) is a clean and generalizable insight. The corrected objective is a reasonable repair, though its cost/benefit relative to the cheaper re-rounding intervention is not fully argued. The census and pre-registration log are valuable. The paper stops short of a complete account (absorption not derived, Mistral unresolved), which is honestly stated but limits the contribution's completeness.

Overall Score

6 / 10 — Weak accept (borderline accept).

The paper identifies a real and important failure mode, explains it with a correct and interpretable theory, supports it with unusually strong causal evidence, and proposes a repair that follows from the theory. It is above the acceptance threshold on contribution and soundness. However, it has three issues that a strict reviewer will weigh: (i) the theory's quantitative predictions are derived under an isotropic approximation that the data only partly supports, and the general-case direction fits worse than the approximation; (ii) the central phenomenon lacks a quantitative collapse criterion despite an exhaustive search; and (iii) the practical recommendation is ambiguous because the cheaper re-rounding intervention is competitive with the proposed repair. If the authors can address (i) with a measurement or a tighter derivation, (ii) with an honest statement of the open problem (which they partly do), and (iii) with a clear argument for when the corrected objective is preferable, the paper would move to a clear accept. As written, I lean accept but would not fight for it in a borderline discussion.

Confidence

4 / 5 — I am familiar with the PTQ literature (GPTQ, OBS/OBC, AWQ, QuaRot, AutoRound, calibration sensitivity) and with the attention-sink/massive-activation literature. I have not independently verified the numerical results, but the internal consistency of the reported values across tables and the pre-registration log increase my confidence in them. I am less confident in the details of the K-FAC / output-channel-factor argument (Proposition 2a), which I believe is correct but which I would want to see spelled out more fully for the general (non-diagonal G) case.

Suggestions for Improvement (ordered by impact)

- 1. Close the isotropic/general gap (W1).** Either measure the anisotropy of M_j at the sink matrices and justify the approximation, or derive the row-concentration and readout signatures directly in the general case. The current post-hoc substitution of v for x_0 is the weakest link in the theory.
- 2. Clarify the status of the row readout (W2).** If it is derivable from Eq. (6), show the derivation; if not, label it as a diagnostic and report the best *a priori* readout from the theory alongside it.
- 3. State the absence of a collapse criterion plainly (W3).** The paper already lists the refuted predictors; add a sentence in the introduction and conclusion that no criterion is currently available and that this is the main open problem.
- 4. Argue the case for the corrected objective over re-rounding (W4).** Add a paragraph in Section 6 comparing the two on cost, robustness, and generality. If re-rounding is the better practical recommendation, say so.
- 5. Add the single-BOS control at the sink matrix (m1) and the λ -held ablation (m4).** Both are small experiments that close obvious reviewer questions.
- 6. Fix the Table 15 inconsistencies (m8).** Correct the model names and reconcile the size range with the text.
- 7. Trim the main text.** Move Tables 15 and 16 to the appendix; promote one summary figure.

8. **Fix the TODO in the reproducibility statement** before submission, or state explicitly that the repository will be released upon acceptance.

Reviewer note: I have not seen the code or the pre-registration log, and my assessment of the empirical results relies on the paper's internal consistency. If the released artifacts contradict any of the reported numbers, my assessment would change accordingly.

What I corrected and what I added

Corrected: the IFEval point range, the quantizer count, the whitened-overlap ranking, the activation-patching location, the "far from one" phrasing, and the table/figure phrasing.

Added: the loop-correction disclosure as a strength (I missed it initially), and the Table 15 inconsistencies (Qwen2.5-34B/3B, Mistral-8B, 0.5B vs 1.7B).

Verified as accurate: all numerical values I cited (A_BOS, G_BOS, IFEval scores, κ values, memory overheads, recall values, Pearson correlations, model counts, protocol counts, pre-registered prediction count, and the specific tables/figures referenced).